

Exam Date & Time: 11-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS -JANUARY 2024

SUBJECT: ECE 3153- MICROPROCESSORS

MICROPROCESSORS [ECE 3153]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Missing data may be suitably assumed.

- 1A) What are the differences between privileged and non-privileged modes in ARM 7? Discuss various operating modes of ARM 7. (4)
- 1B) What is the significance of status flag bits in ARM 7 processor? Explain the function of each flag with bit diagram. (3)
- 1C) Discuss the various methods available for performance improvement in the processor design. (3)
- 2A) Describe the function of the following ARM7 instructions in detail. Assuming the same PRE-condition for all instructions, write the POST conditions of the affected registers and flags for each instruction.
- R0=0x40000000 R1=0x65291A3D R2=0x87341299
R3=0x1A2A3B4C R4=0x12345678 R5=0x00000000
R6=0x45129860 R7=0x88772344 cpsr = NZCvqiFt USER and (4)
0x40005000: 0xAB 0x9C 0x72 0x39
- i) RSC R5, R0, R1, ASR #4
ii) CMN R3, R2, LSL #6
iii) BIC R6, R3, R4, LSR #4
iv) SWPB R7, R4, [R0]
- 2B) The operation code of an ARM 7 instruction is 0xE4118004. Identify the instruction and explain the function. Mention the type of addressing mode employed. (3)
- 2C) Write an embedded C program instruction equivalent to the ARM7 code given below:
- ```
FOO MOV R0, #2
 MOV PC, LR
```
- (3)
- ```
MAIN MOV R1, #3
      BL FOO
      ADD R2, R0, R1
```

- 3A) Write an algorithm and Assembly Language Program in ARM 7 to find the GCD of two 8-bit numbers. Store the result in memory location. (4)
- 3B) Write an assembly language program in ARM 7 to subtract two 32-bit number using 2's complement method. Assume that numbers are stored in the memory and also store the result in memory. Compute and write the result after the execution of each instructions. (3)
- 3C) Which are the directives used to reserve block of memory and also to initialize the reserved memory locations? Describe them with proper syntax and example. (3)
- 4A) A stepper motor and 4 LEDs are interfaced to LPC2148 using P0.0 to P0.3 and P0.4 to P0.7 respectively. Write an embedded C program with comments to explain the logic for rotating the stepper clockwise with full step sequence continuously. Display the corresponding sequence on the LEDs simultaneously. Draw the interface diagram. (4)
- 4B) Discuss the function of the following GPIO registers: IOxPIN, IOxDIR, IOxSET. (3)
- 4C) Write the C statements for the following:
i) set pin 15 of PORT0(i.e. P0.15) as output and drive the pin High.
ii) set pin 11 of PORT0 as output and the output logic '1' on this pin and then to logic 0. (3)
iii) set P0.3 and P0.8 as output and set the logic high.
- 5A) Explain how do you enable and disable IRQ and FIQ interrupts in ARM 7? Write an ARM 7 instructions to perform the following tasks separately:
i) Enable IRQ and FIQ (4)
ii) Disable IRQ and FIQ
- 5B) Discuss the various operating modes and privilege levels in Cortex-M3. With the help of a neat diagram, explain the allowed operating mode transitions. (3)
- 5C) Calculate the bit-band alias address of 4th bit of the bit-band byte present at the address 0x20030300, in Cortex M3. Write the ARM7 instructions to make the specified bit '0' and '1' separately. (3)

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